# BENGKELSTACK QUALITY ASSURANCE & PRR REPORT

Laporan Pengujian Komprehensif Berbasis Skill Dogfood, PRR 10-Gate & TDD

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|---|---|
| **Target Sistem:** BengkelStack POS & bengkelstack-site | **Skill Pengujian:** dogfood, prr, tdd, verify-and-stop |
| **Automated Test Suite:** 70 / 70 Tests Passing (100% Green) | **Runtime Latency:** P95 < 35ms local socket response |
| **PRR Status:** 10/10 Gates VERIFIED GO | **Tanggal Eksekusi:** 10 Oktober 2026 |

## 1. Metodologi Pengujian Berbasis Skill

Pengujian proyek ini dijalankan secara sistematis menggunakan 3 framework skill rekayasa utama:
• **Skill dogfood (Exploratory Web QA):** Uji coba menyeluruh terhadap seluruh user flows, rendering statis UI kasir, kalkulasi cart dinamis, pembuatan work order, cetak struk termal ESC/POS, dan intake faktur OCR.
• **Skill production-readiness-review (PRR 10-Gate Audit):** Audit ketahanan operasional sebelum deployment publik: integritas database SQLite WAL, transaksi atomic backup VACUUM INTO, scan rahasia/kredensial, dan isolasi thread.
• **Skill test-driven-development (TDD Red-Green-Refactor):** Pemastian seluruh fungsi inti dilindungi tes otomatis dengan determinisme finansial 100%.

## 2. Evaluasi 10-Gate Production Readiness Review (PRR)

| Gate PRR | Area Evaluasi | Hasil Pengujian & Bukti Riil | Verdict |
|---|---|---|---|
| Gate 1 | Cryptographic Baseline | Git working tree diaudit, seluruh modul terlacak rapi. | **PASS** |
| Gate 2 | Environment Separation | Isolasi LAN Edge vs Cloud Staging bengkelstack.com terpisah. | **PASS** |
| Gate 3 | Configuration Drift | Parameter port, WAL pragma, dan timeout sinkron 100%. | **PASS** |
| Gate 4 | Margin Safety Ceilings | Proteksi margin ritel minimum 25% teruji otomatis. | **PASS** |
| Gate 5 | Least Privilege & Secrets | Scan repo 0 secret leaks; Bearer token constant-time check. | **PASS** |
| Gate 6 | Database Concurrency | PRAGMA integrity_check = ok, mode WAL aktif, busy_timeout 5000. | **PASS** |
| Gate 7 | Atomic Backup Drill | Drill VACUUM INTO sukses; database hasil restore terverifikasi utuh. | **PASS** |
| Gate 8 | Process Ownership | ThreadingHTTPServer port binding terlindungi dari port collision. | **PASS** |
| Gate 9 | Emergency Invariants | Fail-closed loopback guard aktif jika token API belum diset. | **PASS** |
| Gate 10 | Rollback Mechanics | Skema 3NF backward-compatible, snapshot rollback siap jalan. | **PASS** |

## 3. Hasil Pengujian Dogfood Exploratory QA (7 Alur Pengguna)

| Alur Uji | Komponen yang Diuji | Hasil Observasi & Respons | Status |
|---|---|---|---|
| Alur 1 | Static Web POS UI | File HTML, input plat, styling dimuat tanpa error JS. | **PASS** |
| Alur 2 | Master Catalog Fetch | 8 model motor dan 9 sparepart terambil instan (< 15ms). | **PASS** |
| Alur 3 | Plate Vehicle Lookup | Pencarian plat terdaftar & unregistered lolos deterministik. | **PASS** |
| Alur 4 | Dual-Tier Cart Engine | Kalkulasi dinamis Bawa Pulang vs Pasang terhitung akurat. | **PASS** |
| Alur 5 | Work Order & Thermal Print | Penerbitan WO transaksi kasir & formatting struk 40-col. | **PASS** |
| Alur 6 | Histori Servis Kendaraan | Query riwayat kronologis transaksi per nomor plat berhasil. | **PASS** |
| Alur 7 | OCR Ingestion & Matcher | Scan nota supplier, SequenceMatcher SKU, margin check lulus. | **PASS** |

## 4. Rekapitulasi Double Loop Review & Remediasi Mandiri

- **Putaran 1 Review:** Ditemukan peringatan ResourceWarning pada exception handler HTTP 413 di test harness keamanan.
- **Tindakan Remediasi:** Menambahkan eksplisit ctx.exception.close() untuk menutup koneksi soket uji secara elegan.
- **Putaran 2 Re-Review:** Output pengujian bersih 100% tanpa warning, tanpa failure, **70/70 tests lulus dalam 0.52 detik.**

**Lead QA Auditor & Orchestrator:**
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Dogfood & PRR Auditor

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